

2023-Ph-H1-Q14

Section: Particles and Waves

Topic: Radiation

Which radiation has no mass and no charge?

Answer: C

(Gamma)

Guidance for Students:

Gamma rays are high-energy electromagnetic waves. Unlike alpha (He nucleus) or beta (electron), gamma has no mass or charge.

Revision Tips:

- Alpha: 2 protons + 2 neutrons → heavy, +2 charge.
- Beta: electron or positron → small mass, ± 1 charge.
- Gamma: just energy → no mass, no charge.